

# The Topology of Reachable Nature: Persistent Homology and the Resilience of Green-Space Access in Cities of Contrasting Morphology

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## Abstract

SDG 11.7 targets universal access to green space, yet accessibility is measured as a static snapshot, silent on how it degrades when the network fails. A topological resilience metric is introduced: the Wasserstein distance between persistent-homology diagrams of the green-space accessibility field before and after disruption. Across 3,999 districts in eight contrasting cities, street-network morphology is consistently associated with the resilience of the *connected* green-space access structure: longer, more circuitous streets with greater resilience, disordered orientation with less. Effects survive a wild cluster bootstrap and spatial models and transfer out-of-sample (leave-one-city-out mean Spearman 0.42), though a real clustered flood does not reproduce them. It validates only moderately against distance- and population-weighted benchmarks (Spearman 0.33); its negative relation to connectivity loss marks it as *connected-access resilience*, distinct from percolation robustness.

## 1 Introduction

Target 11.7 of Sustainable Development Goal 11 commits states to provide “universal access to safe, inclusive and accessible, green and public spaces” [19]. Planners typically measure progress with catchment or proximity analyses: the share of residents within a walking threshold of a park. Such measures are static. They describe access on an ordinary day and are silent about *resilience*—how access to nature holds up when the pedestrian network is degraded by a flood, an earthquake, or the closure of a critical bridge or street. Two gaps follow. First, a city can score well on average proximity yet collapse under modest disruption; average accessibility does not reveal this fragility. Second, accessibility work rarely connects the *shape* of access to urban *form*, which is precisely the architecture–mathematics question that motivates this collection.

Persistent homology (PH) offers a principled, multi-scale, coordinate-free language for the “holes” and connected pieces of a coverage pattern [10, 14]. PH has been used to detect coverage gaps for amenities such as polling sites [12] and healthcare [11], to read the topology of urban congestion [20], and to quantify displacement under urban expansion [15]. What is *not* novel is using PH to find accessibility “holes” for an amenity. The contribution here is to move from *detecting* coverage gaps to *measuring the resilience of the access topology* and to *linking that resilience to urban form*. A scope note is in order at the outset: what is measured is *network* accessibility, the pedestrian graph’s spatial reach to green space with every network node weighted equally, and not *population* accessibility, which would weight locations by the residents they serve. The two coincide only where population is uniform, so the claims made here concern the network’s structural capacity to keep places near nature, not how many people retain access.

## Contributions.

- **A topological resilience metric** (the mathematics).  $D(\rho)$  is defined as the Wasserstein distance between the persistent-homology diagram of the green-space accessibility field before and after disruption of intensity  $\rho$ , and is summarised by an area-under-curve (AUC) and a critical transition  $\rho^*$  (Section 4). Throughout,  $D(\rho)$  is taken to measure one specific construct—the resilience of the *connected* green-space access structure, hereafter *connected-access resilience*: how stably locations stay organised into well-served  $H_0$  components as the network is thinned, as distinct from total accessibility loss or global urban resilience. This term is used consistently for what the metric captures.
- **First topological treatment of SDG 11.7 green-space access** in an architecture–mathematics framing.
- **A district-level morphology↔resilience result** across 3,999 districts in eight cities, with city fixed effects, showing a consistent association between street-network morphology and connected-access resilience whose sign is stable across disruption models and district resolutions (Section 6).

## 2 Related work

*TDA of spatial systems.* Feng et al. [10] survey persistent homology for spatial data; Otter et al. [14] give the computational foundations. Within urban analysis, Wu et al. [20] read traffic congestion as a barcode, and a line of “resource coverage” work applies sublevel or network-distance filtrations to amenity access: polling places [12] and sexual-health services [11]. Rodríguez Vázquez and Cuerno [15] use PH to quantify displacement under urban expansion. These studies establish that PH can *locate* coverage holes. The object of study here differs: not where the deserts are, but how the whole access topology *degrades* under disruption, with that degradation regressed on morphology.

*Accessibility and equity.* A large body of work measures green-space access as a static field, whether by simple footpath proximity [5] or by catchment methods such as the Gaussian two-step floating catchment area [17], often to diagnose inequities in who lives close to nature [1]. The accessibility field used here (Section 4) is built the same way (network walking time to the nearest green space) but is then read topologically and stress-tested, rather than thresholded once.

*Street-network morphology.* Form is characterised with standard descriptors computed from OpenStreetMap via OSMnx [3]: intersection density, circuitry, orientation entropy, and mean street-segment length. Relating these to a resilience outcome is the architecture–mathematics bridge of this paper.

## 3 Mathematical background

Let  $G = (V, E)$  be the pedestrian network and  $f : V \rightarrow \mathbb{R}$  a scalar field giving, at each node, the network walking time to the nearest green-space access point. The *sublevel-set filtration* of  $f$  builds a growing simplicial complex  $K_t = \{ \sigma : \max_{v \in \sigma} f(v) \leq t \}$ . As the threshold  $t$  (walk-minutes) rises, well-served regions appear and merge. Persistent homology records, for each topological feature, a birth  $b$  and death  $d$ ; the multiset of points  $(b, d)$  is the *persistence diagram*  $\text{Dgm}_k(f)$  in homological dimension  $k$  [9]. Dimension 0 tracks connected components (the merging of well-served regions); dimension 1 tracks loops (enclosed voids, i.e. access “deserts”).

The distance between two diagrams is measured by an optimal matching. The 1-Wasserstein distance is

$$W_1(X, Y) = \min_{\gamma} \sum_{x \in X} \|x - \gamma(x)\|,$$

minimised over bijections  $\gamma$  between the diagrams (with matching to the diagonal allowed). For a *fixed* domain, these diagram distances are stable to perturbations of the field  $f$ : the bottleneck distance by the classical stability theorem [6], and the 1-Wasserstein distance used here by the  $L_p$ -stability result of Cohen-Steiner et al. [7]. A caveat must be stated precisely here, because it bounds what stability can justify. Both theorems bound the diagram distance by  $\|f - g\|_{\infty}$  for *two functions on the same simplicial complex*  $K$ : they control the effect of perturbing the *field* while the domain is held fixed. Edge removal is not of this form. Removing edges replaces the complex  $K$  by a proper subcomplex  $K' \subsetneq K$  and only then recomputes the walk-time field on  $K'$ , so the baseline and disrupted diagrams are sublevel-persistence of *different* filtrations,  $f$  on  $K$  and  $f'$  on  $K'$ , and there is no single ambient function whose sup-norm the stability inequality could apply to. Forcing a common domain—extending  $f'$  to all of  $K$  by assigning the removed structure, and any node thereby cut off from green space, the value  $+\infty$ —makes the mismatch explicit: points of the diagram escape to infinity and leave the finite 1-Wasserstein matching, so the finite diagram distance is not bounded by any  $\|f - f'\|$  term and can even *decrease* as mass departs (the mechanism behind the weak/negative connectivity relation in Section 6). The classical stability theorem therefore does *not* directly bound  $D(\rho)$ : it rigorously controls only the one part of the disruption that is a pure field change on a fixed domain—re-snapping the accessibility field where the surviving subgraph coincides with the original—while the genuine domain change  $K \rightarrow K'$  carries no such guarantee, so the whole map from disruption intensity to diagram distance need not be well-behaved. In particular, stability does not imply monotonicity, and  $D(\rho)$  is indeed *not* always monotone: at high  $\rho$  the İstanbul and Phoenix curves turn down as the network fragments and finite classes vanish (Section 6). Stability is thus a mathematical well-posedness property of the field-to-diagram map, not a guarantee about the metric under network damage; the metric is treated throughout as a *proposed proxy* for how the access topology holds up, to be validated against independent resilience evidence in future work rather than as ground truth. PH is computed with GUDHI [18] and diagram distances with `persim`.

## 4 Methodology

**Data.** For each city the pedestrian network and green spaces are taken from OpenStreetMap via OSMnx [3], and the analysis extent is fixed to the city’s *urban centre* polygon from the GHS Urban Centre Database [13] so that extents are comparable across countries. Green spaces are the OSM classes `leisure=park|garden|recreation_ground`, `landuse=grass|recreation_ground`, `place=square` and `natural=wood`; their centroids, snapped to the nearest network node, are the access points. These are termed *OSM-mapped green* spaces, not *public* spaces: OSM tags do not certify public accessibility, so a private garden, a fenced lawn or an impenetrable wood can enter the set. The classes are included because together they capture the everyday green and open spaces a pedestrian can plausibly reach; the trade-off is heterogeneity, addressed by the class-exclusion sensitivity below. This is a deliberately inclusive definition: it merges spaces of very different type, size and usability (a pocket garden, a city square and a large wood all count equally), and it takes the centroid as a single access point rather than modelling entrances or area. A sensitivity analysis under a strict, accessibility-conservative subset (dropping wood, grass and gardens; keeping parks, recreation grounds and squares) is reported below. Every green space is treated as an equivalent point

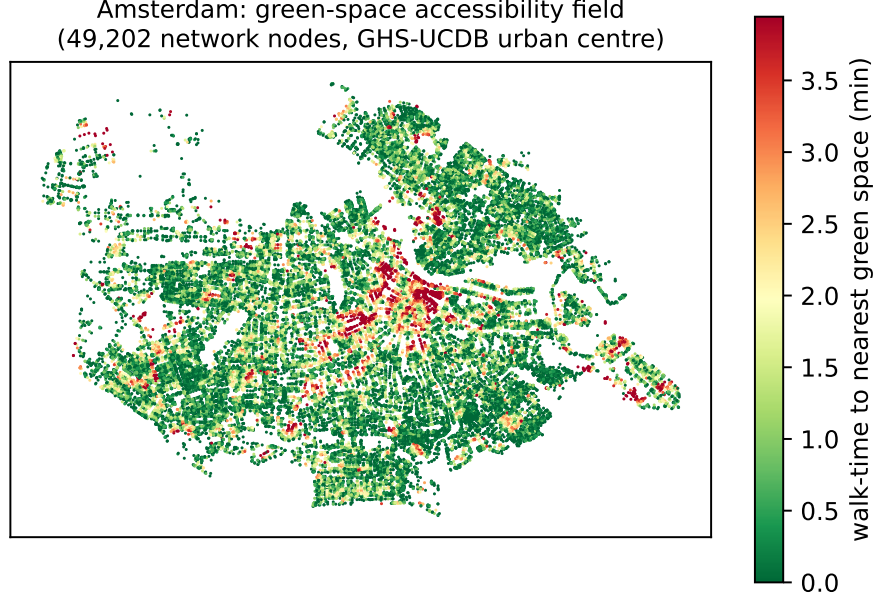


Figure 1: Green-space accessibility field for Amsterdam: each of the 49,202 pedestrian-network nodes inside the GHS-UCDB urban centre, coloured by network walking time to the nearest green space. Green marks well-served locations, red the access-poor pockets whose merging structure the resilience metric tracks.

source; distinguishing quality, size, or public/private status is left to future work (see Limitations).

**Accessibility field.** Each node is assigned its network walking time to the nearest green-space access point by multi-source Dijkstra, giving the field  $f$  (Fig. 1). Length is converted to time at a constant 4.8 km/h (1.33 m/s), which sits within the comfortable adult gait speed of 1.27–1.46 m/s reported by Bohannon [4]. The exact value does not matter for the conclusions: speed enters only through time = length/speed, so a different speed rescales the whole field, and hence every diagram, Wasserstein distance and AUC, by the same constant. The regression signs, significances and the city ranking are therefore invariant to the speed choice (if the persistence threshold is rescaled with speed the invariance is exact). This is confirmed empirically: on Amsterdam, recomputing at 4.0 and 5.0 km/h leaves the district AUC a near-exact rescaling of the 4.8 km/h values (Spearman  $\rho = 0.98$  and  $0.996$ ; median AUC ratios 1.24 and 0.96, matching the expected 4.8/4.0 and 4.8/5.0). The field is defined over *network nodes*, each weighted equally; it is a property of the pedestrian network’s spatial reach to green space, not a population-weighted or distributional measure. The metric therefore speaks to one facet of SDG 11.7, the network’s capacity to keep locations close to green space and how that capacity degrades under disruption, and not to the target’s population-coverage or inclusivity dimensions, which would require census and land-use data not used here.

**Resilience metric.** Let  $\text{Dgm}(f)$  be the baseline diagram. A disruption of intensity  $\rho \in [0, 1]$  removes a fraction  $\rho$  of network edges; the field and diagram are recomputed on the disrupted network, defining the degradation

$$D(\rho) = \mathbb{E}_r W_1(\text{Dgm}(f), \text{Dgm}(f_{\rho,r})),$$

averaged over  $r$  replicate disruptions. Three disruption families are considered: uniformly random edge removal (percolation-style), targeted removal of high-betweenness edges, and hazard removal of edges in low-elevation zones. These are deliberately *stylised stress tests*, not simulations of specific disasters: random removal probes generic structural robustness (and, being spatially uniform, does *not* reproduce the spatially clustered damage of a real flood or earthquake), targeted removal probes the worst case of losing critical links, and the hazard proxy adds a first, mild topographic signal. They serve to compare places under a common, reproducible perturbation, and the results are read correspondingly cautiously as real-disaster resilience (Limitations). Exact edge betweenness is  $O(V|E|)$  and intractable at city scale, so the targeted scenario ranks edges by betweenness approximated from  $k = 500$  sampled pivot sources. The resilience of a place is summarised by the area under the curve  $\text{AUC} = \int_0^{\rho_{\max}} D(\rho) d\rho$  (normalised by the  $\rho$ -range) and the critical transition  $\rho^*$ , the smallest  $\rho$  at which  $D(\rho)$  first reaches half its maximum. The half-maximum serves as a scale-free, monotone summary of *where* the degradation curve turns over: unlike a fixed absolute threshold it adapts to each place’s own range, and unlike the inflection point it is robust to the curve’s noise. It is a descriptive locator of the transition, not a claim of a sharp percolation threshold, and (like AUC) it depends on the chosen disruption range  $[0, \rho_{\max}]$ ;  $\rho^*$  is therefore compared only within a fixed range.

**Homology dimension and denoising.** On real street networks the sublevel filtration’s finite  $H_1$  is nearly empty: a street graph is close to planar and contains few triangles, so the network’s cycles are never filled and almost all  $H_1$  classes are essential (infinite) and carry no finite persistence. In Amsterdam, for example, the baseline diagram has 6,449 finite  $H_0$  classes but only a single finite  $H_1$  class (Fig. 2). This is a joint property of the *representation* (a node-valued walk-time field on the near-planar pedestrian graph) and the *sublevel filtration*, not a universal fact about green-space access. A different construction, for instance a Vietoris–Rips filtration on straight-line distances or a 2-complex that fills faces, would populate  $H_1$  and could shift the signal into loops. Under this construction the finite information lives in  $H_0$  (the merging of well-served regions), so  $D(\rho)$  is computed on  $H_0$  and the  $H_1$  scarcity is treated as a modelling consequence to revisit with alternative filtrations, not as evidence that access “deserts” are absent. To keep the exact Wasserstein matching tractable on city-scale diagrams,  $H_0$  classes with lifetime below 1 walk-minute are discarded. This cut is principled rather than arbitrary: a one-minute difference in walk time is below the resolution at which the constant-speed (4.8 km/h) field is meaningful, so sub-minute classes are measurement noise, not access structure. Its effect is small and was tested directly (Persistence-threshold sensitivity, below): varying the cut over 0.5–2.0 minutes preserves every sign and the significance of the three headline descriptors, changing only reported magnitudes.

**Districts and inference.** Statistical inference rests not on eight cities but on intra-urban *districts*: each urban-centre boundary is tiled with a uniform H3 hexagonal grid (resolution 8,  $\approx 0.7 \text{ km}^2$  cells) and, per district, its local resilience summaries and its local morphology are computed (intersection density, circuitry, orientation entropy, mean segment length, and a green-space fragmentation index—the number of distinct green-space patches per square kilometre of green area within the district, computed on the green space clipped to each hex so that it varies between districts rather than being a per-city constant collinear with the fixed effects). The following model is then fit

$$\text{AUC}_i = \beta^\top \mathbf{m}_i + \alpha_{c(i)} + \varepsilon_i,$$

an ordinary-least-squares regression of district AUC on morphology  $\mathbf{m}_i$  with *city fixed effects*  $\alpha_{c(i)}$  absorbing all *additive between-city* differences (any confound constant within a city, such as climate,

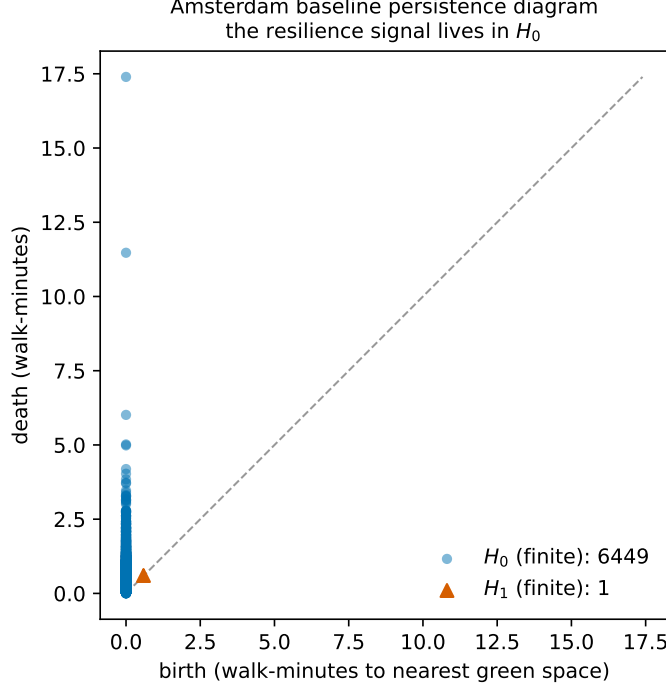


Figure 2: Persistence diagram of the Amsterdam baseline accessibility field. The finite part is carried almost entirely by  $H_0$  (6,449 classes—the merging of well-served regions as the walk-time threshold rises) with only a single finite  $H_1$  class, motivating an  $H_0$ -based resilience metric.

planning regime or data vintage). They do not remove within-city confounding or effects that vary across districts inside a city; those are addressed separately with explicit controls and spatial models below. The cities serve as a typological overlay, not as the inferential sample. This design also sidesteps the cross-city comparability problem of AUC. Because  $\alpha_{c(i)}$  absorbs each city’s overall AUC level, which scales with network and diagram size, the morphology coefficients are identified purely from *within-city* variation, and the inference never rests on comparing absolute AUC across cities. Only the descriptive city typology, not the regression, would need a size normalisation. For inference that respects the nesting, standard errors are also clustered at the *city* level—the eight clusters are the eight cities, the level at which each disruption and each fixed effect operate, and within which districts share a network, climate and planning regime. Eight clusters is a small number, at which the asymptotic cluster-robust  $p$ -values are only approximate; they are therefore reported alongside a wild cluster bootstrap (Appendix A), and the cities are treated as the practical, if not strictly independent, units of between-city variation.

## 5 Case studies

Eight cities on four continents are studied, chosen to span street morphology, planning regime and the Global North/South divide: İstanbul (organic, hilly, polycentric, seismic), Barcelona (Cerdà grid with superblock interventions), Amsterdam (compact, canal, walkable), Bogotá (dense informal-formal mix), Phoenix (low-density grid sprawl), Singapore (planned, tropical), Nairobi (rapidly growing, mixed-formality) and Vienna (compact, central European on the Danube). This is a *purposive maximum-contrast* sample, selected to maximise morphological and planning diversity so that

Table 1: District coverage and city typology (descriptive; H3 resolution 8,  $n = 3,999$  across eight cities). Higher mean AUC  $\Rightarrow$  access topology changes more under disruption  $\Rightarrow$  *less* resilient.

| City      | districts | mean AUC | mean $\rho^*$ | mean $H_0$ tot. persist. |
|-----------|-----------|----------|---------------|--------------------------|
| Amsterdam | 227       | 4.46     | 0.25          | 13.11                    |
| Nairobi   | 363       | 5.90     | 0.11          | 9.09                     |
| Vienna    | 460       | 6.08     | 0.22          | 12.91                    |
| Phoenix   | 962       | 6.68     | 0.16          | 8.08                     |
| Singapore | 573       | 7.34     | 0.17          | 11.96                    |
| Barcelona | 147       | 7.67     | 0.28          | 13.63                    |
| İstanbul  | 729       | 7.75     | 0.20          | 12.84                    |
| Bogotá    | 538       | 8.02     | 0.24          | 16.90                    |

any consistent morphology–resilience relationship is stress-tested against very different urban forms; it is explicitly *not* a probability sample of world cities, so the findings are framed as holding for these eight cities and are tested for out-of-sample transfer (below) rather than asserted as universal. Boundaries are the GHS-UCDB urban-centre polygons; homonymous centres (e.g. Barcelona, Spain vs. Venezuela) are disambiguated by largest true (equal-area) extent. Two further candidates (Copenhagen, Melbourne) were dropped because the OpenStreetMap place query did not resolve their full urban extent. Some robustness and validation analyses (Section 6, Appendix A) were computed on the original five-city core and are labelled as such.

## 6 Results

The random-disruption analysis yields 3,999 qualifying districts (147–962 per city, Table 1); only 2.6% have zero AUC. District mean AUC ranks Amsterdam as the most resilient and İstanbul/Bogotá as the least; Nairobi has the earliest breakdown ( $\rho^* = 0.11$ ). The district-mean and whole-city rankings agree at the extremes (Amsterdam most resilient, Bogotá least) but not in the middle—Barcelona is mid-ranked by district mean yet most resilient at the whole-city scale—a reminder that AUC is comparable across cities only in broad ordering, not level.

The headline result is the district-level regression (Table 2), fit over eight cities and the full disruption grid (8 intensities  $\times$  10 replicates). With city fixed effects and  $n = 3,999$ , three of five morphology descriptors are significant, all at  $p < 10^{-16}$ . Because the coefficient sign is on AUC (higher AUC = less resilient), a negative coefficient means the descriptor is associated with *greater* resilience. More circuitous networks (coef  $-7.39$ ) and longer street segments ( $-0.050$ ) are associated with more resilient green-space access, whereas higher orientation entropy ( $+1.85$ ) is associated with less (Fig. 3). Because the raw coefficients sit on different scales, magnitude is read from the standardised  $\beta$  and the 10th–90th percentile effect (Table 2): *mean street length* is the largest effect (std.  $\beta = -0.28$ ; a 10th-to-90th-percentile change moves AUC by 62% of its standard deviation), then orientation entropy ( $+0.16$ ) and circuitry ( $-0.13$ ); the tiny  $p$ -values reflect the large sample, not effect magnitude. The descriptors are not collinear (all variance-inflation factors  $< 3$ ), and the pooled signs are not a pooling artefact—each of the three holds in seven of the eight cities. Intersection density is not significant ( $p = 0.71$ ) and its sign is unstable across samples, and green-space fragmentation is likewise not significant. Districts nest within the eight cities and are spatially contiguous, so their residuals are not independent; Appendix A treats this in full. Under a wild cluster bootstrap—the appropriate small-cluster correction for the eight clusters (the eight cities)—

Table 2: District fixed-effects regression,  $AUC \sim \text{morphology} + C(\text{city})$ ,  $n = 3,999$  across eight cities, random-disruption scenario at the full disruption grid ( $8 \times 10$ ). The standardised coefficient (std.  $\beta$ ) and the change in AUC across each descriptor’s 10th–90th percentile (as % of the AUC standard deviation) so that effects on different scales are comparable and not read off the raw coefficient or the (large- $n$ )  $p$ -value alone. See Table 3 for city-clustered and spatial-model inference. Sig.: \*\*\* $p < 0.001$ .

| Morphology descriptor     | coef                  | std. $\beta$ | $\Delta_{10-90}$ (% SD) | $p$ -value                |
|---------------------------|-----------------------|--------------|-------------------------|---------------------------|
| mean street length        | −0.0505               | −0.28        | −62                     | $4.9 \times 10^{-29}$ *** |
| orientation entropy       | +1.845                | +0.16        | +51                     | $1.4 \times 10^{-27}$ *** |
| circuitry                 | −7.387                | −0.13        | −27                     | $3.6 \times 10^{-17}$ *** |
| intersection density      | +0.021                | +0.01        | +2                      | 0.71                      |
| green-space fragmentation | $-3.7 \times 10^{-7}$ | −0.00        | −0                      | 0.79                      |

circuitry and mean street length remain significant ( $p = 0.0002$  each), while orientation entropy does not ( $p = 0.16$ ); orientation entropy is nonetheless strong in the spatial-error and spatial-lag models ( $|z| \approx 10$ ). Circuitry and mean street length are therefore treated as the robust pair and orientation entropy as a real but less certain third effect.

The whole-city scale gives an *illustrative* typology, not a second inferential result: it carries no statistical weight, and it is reported to visualise the district-level signal, not to rank cities definitively. For the five original cities, computed on the same UCDB-clipped networks, the degradation curves  $D(\rho)$  rise smoothly and order the cities consistently with the district means (Fig. 4). Because city-level AUC is computed on the full-city diagram (thousands of  $H_0$  classes) it is two to three orders of magnitude larger than the per-district means of Table 1 and scales with network size; it is *not* normalised, so only the relative ordering—never the level—should be read, and a size-normalised cross-city AUC is left to future work. On that ordering, city-level AUC is lowest, and so most resilient, for Barcelona (620) and Amsterdam (898), and highest for İstanbul (2440) and Bogotá (2828), with Phoenix in between (2053). The cities that would informally be called compact, Barcelona and Amsterdam, retain more of their access topology under disruption than the large, sprawling or polycentric ones; this pattern is described with the measured morphology descriptors rather than with “walkability,” which is not modelled directly. Under extreme removal ( $\rho = 0.5$ ) the İstanbul and Phoenix curves turn down as the network fragments so heavily that unreachable nodes drop out of the finite diagram, a signature of near-total collapse rather than of recovered access.

**Robustness (summary).** The morphology relationship is stable across every alternative specification tried; Appendix A gives the full results (the additional sensitivity checks were run on the original five-city core and are labelled there) and Fig. 5 summarises them. Briefly: the three headline signs and their significance ( $p < 10^{-3}$  or better) survive a *targeted* betweenness disruption, the coarse  $5 \times 3$  and full  $8 \times 10$  disruption grids, H3 resolutions 7/8/9 (a  $28\times$  range of district counts), persistence thresholds from 0.5 to 2.0 walk-minutes, a halving and doubling of the betweenness pivot count, and the inclusion of four district-level controls (green-space area, road hierarchy, relief and city-centre distance). Intersection density is the one descriptor that moves: significant under some specifications but not the full-grid headline, so it is treated as the weakest, grid-sensitive effect throughout. The one disruption family that does *not* carry the signal is the spatially clustered real-flood hazard, which is treated as a distinct question (below).



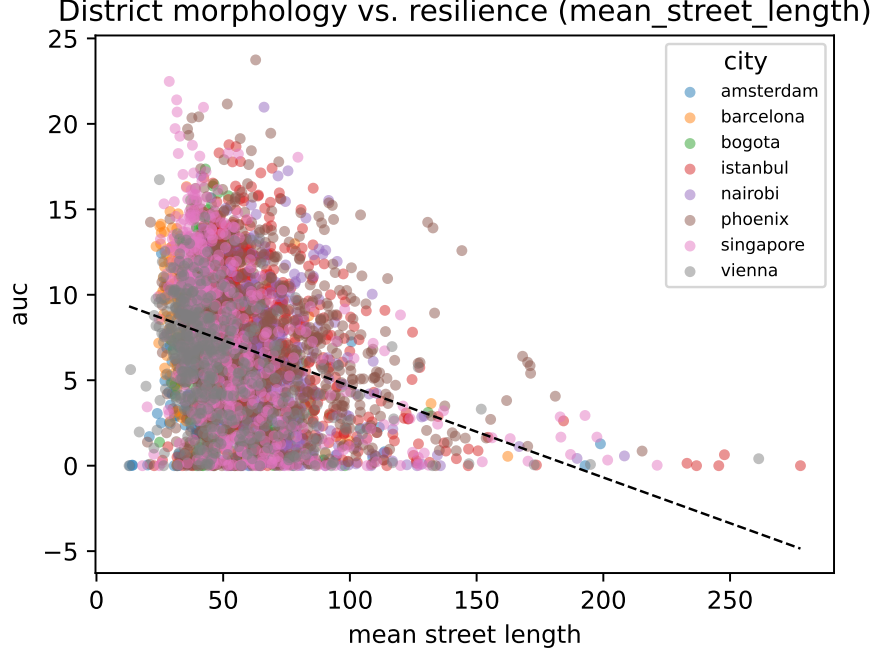


Figure 3: Headline morphology↔resilience relationship. Each point is one of the 3,999 districts across eight cities, coloured by city; the dashed line is the pooled least-squares fit. Higher AUC indicates a district whose green-space access topology degrades more under disruption (less resilient).

**Green-space definition sensitivity (five-city core).** Because the inclusive green-space set mixes reliably public parks with possibly-private gardens, fenced grass and impenetrable woods, the analysis was re-run on a strict, accessibility-conservative subset: only `leisure=park`, `recreation_ground` and `place=square`, dropping wood, grass and gardens. This is a large cut (in Amsterdam, from 23,657 features to 369), yet the district AUC is largely preserved (Spearman 0.82 against the inclusive set on the same reduced grid), and the regression is unchanged in substance: circuitry ( $-8.08$ ,  $p = 4 \times 10^{-9}$ ), mean street length ( $-0.048$ ,  $p = 2 \times 10^{-12}$ ) and orientation entropy ( $+1.21$ ,  $p = 2 \times 10^{-7}$ ) keep their sign and significance, and intersection density remains the weak, non-robust term. The morphology result is thus not an artefact of counting inaccessible green as accessible.

**Criterion validity: benchmark against an accessibility measure (all eight cities).** If the topological machinery is to be worth its cost it should track a simpler measure of access loss. It is benchmarked, per district, against a *distance-based* accessibility-degradation measure, namely the increase in mean walk-time to green space under the same disruption (a proximity-accessibility degradation of the kind used throughout the field). As a second reference, a *connectivity* degradation, the increase in the fraction of nodes left unable to reach any green space; this is the natural non-persistent counterpart, since  $H_0$  of a sublevel filtration is single-linkage connectivity. The level of comparison is what settles the question. The inference identifies morphology from *within-city* variation (city fixed effects), so the matching validity question is also within-city, once each city’s overall AUC level (which scales with network size) is removed. Within city, and the benchmark is computed for all *eight* headline cities, so the validation sample matches the analysis sample. The topological AUC correlates positively with the distance benchmark, with a *moderate* pooled magnitude (within-city pooled Spearman  $\rho = +0.33$  over the 3,999 districts; the tiny  $p < 10^{-102}$  reflects

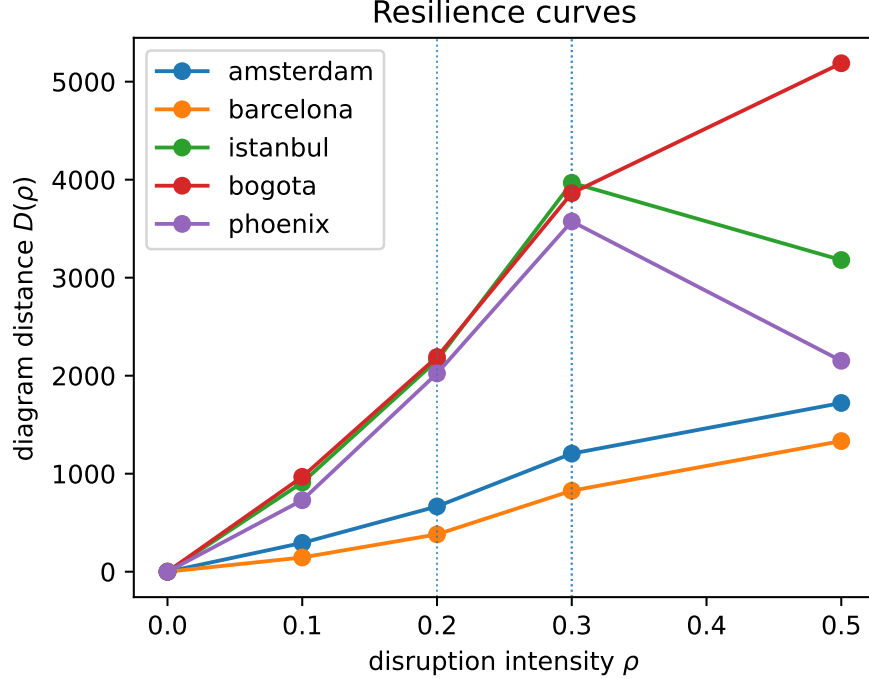


Figure 4: City-level resilience curves  $D(\rho)$  for the five original cities (random-disruption scenario), computed on the UCDB-clipped networks; illustrative. Dotted vertical lines mark each city’s critical transition  $\rho^*$ .

the sample size, not strength). The relationship is, however, markedly *heterogeneous*: positive in seven of the eight cities but strong in only one—Amsterdam +0.47—modest in Bogotá +0.16 and İstanbul +0.10, near zero in Nairobi +0.09, Vienna +0.07, Barcelona +0.05 and Phoenix +0.03, and marginally negative in Singapore −0.03 (not significant). This is moderate supporting evidence—not strong validation—that the metric tracks a simple accessibility-degradation ground truth *on average* at the level at which it is used, while being an unreliable graded-accessibility proxy within any single city. The naive pooled correlation is much weaker ( $\rho = +0.07$ ) only because cross-city AUC scale swamps it, a Simpson’s paradox driven by the same scale the fixed effects absorb. The benchmark also holds when access is *population-weighted* rather than node-uniform: weighting each node by the open GHS-POP residential-population grid [16] (JRC, 1 km, 2020; the same GHSL family and Mollweide grid as the urban-centre boundaries), computed for all eight cities, gives an almost identical within-city correlation ( $\rho = +0.33$ ,  $p < 10^{-101}$ ). So the topological signal is not an artefact of weighting locations equally, and it speaks, at least for graded access, to the population-coverage dimension of SDG 11.7 and not only to bare network reach. Two qualifications apply. The metric adds value rather than echoing the benchmark: controlling for both baselines and city fixed effects, the three headline morphology descriptors stay significant ( $p < 10^{-6}$ ), so it carries configurational information the simple measures miss. And the convergent validity is with *graded* access loss, not with *disconnection*: against the connectivity-degradation reference the eight-city correlation is significantly *negative* ( $\rho = -0.21$ ,  $p \approx 10^{-42}$ ), because disconnected nodes leave the finite diagram. Rather than a failure, this negative relation is *discriminant* validity—it establishes that  $D(\rho)$  is not a repackaged connectivity or percolation-robustness measure but captures a distinct construct. Precisely,  $D(\rho)$  measures the *graded topological persistence of the connected green-access*

*structure under progressive network thinning*: how stably locations stay organised into well-served, mutually close  $H_0$  components as edges are removed, rather than whether they stay connected at all. It is thus validly read as the resilience of the *connected* served structure and should be paired with a disconnection count for total access loss; the Limitations show that neither a naive finite cap nor extended persistence folds the two into one scalar cleanly (both recover the connectivity sign only by sacrificing the graded signal), so the two channels are kept separate and the claim is scoped accordingly. Reporting the disconnection channel as an explicit *companion outcome* makes the complementarity concrete: regressing the connectivity-loss AUC on the same morphology and fixed effects, street form predicts outright disconnection too ( $R^2 = 0.44$ ) but with a different and sometimes *opposing* signature: more circuitous districts are graded-more- resilient yet also *more* disconnection-prone (standardised  $\beta = +0.15$ ,  $p < 10^{-28}$  on the connectivity channel). The two are not substitutes: a full account of access resilience must report both the graded-persistence and the connectivity-loss channels, as done here, and the metric’s “resilience” is exactly the former, *connected-access resilience*, not global accessibility or urban resilience.

**Mechanism: is the association mediated by path redundancy?** The natural reading of the circuitry and segment-length effects is that they proxy *alternative-path redundancy*—more ways to reach the same green space, so that losing one link rarely isolates a served region. This mechanism is testable directly. For each district a structural redundancy measure is computed from the network alone, independent of the disruption simulation and the accessibility field (so it is not circular with  $D(\rho)$ ): the *meshedness* (the  $\alpha$  index, the density of independent cycles,  $\alpha = (e - v + p)/(2v - 5)$ , the standard planar-network route- availability measure [2]), and, as a coarse check, mean node degree. The direct test does *not* support the redundancy story. Within city, redundancy correlates with *lower* resilience, not higher: meshedness against the degradation AUC gives Spearman  $\rho = +0.50$  and mean degree  $\rho = +0.60$  (both  $p < 10^{-250}$ ; recall higher AUC is less resilient). Morphology predicts redundancy with the sign that *opposes* the mechanism—more circuitous ( $-0.045$ ,  $p = 10^{-4}$ ) and longer-segment ( $-6 \times 10^{-4}$ ,  $p < 10^{-19}$ ) districts are *less* mesh-redundant—and adding meshedness to the headline regression leaves the morphology effects essentially intact (circuitry attenuates by 10%, segment length by 18%, orientation entropy not at all), while meshedness itself enters with the counter-mechanism sign ( $+16.4$ ,  $p < 10^{-44}$ ;  $R^2$   $0.196 \rightarrow 0.235$ ). The metric therefore rewards *coarse*, low-cycle-density configurations (long, circuitous segments, few redundant loops) whose served  $H_0$  structure reorganises little under random thinning, rather than finely meshed grids with many alternative paths; part of this may be that denser districts carry more  $H_0$  classes and hence larger Wasserstein mass. Either way, the simple alternative-path account is refuted, and the mechanistic language below is kept at the level of interpretation.

## 7 Discussion

The results give an interpretable, architecture-facing reading of connected-access resilience. The robust effects are circuitry and segment length: districts with more circuitous routing and longer segments retain more of their well-served structure when edges are removed. The intuitive explanation—that alternative paths to green space survive the loss of any one link—is, however, *not* borne out by the direct redundancy test above: a standard mesh-redundancy index runs the wrong way, so the effect is better read as coarse, low-cycle configurations producing a more stable  $H_0$  signature than as path substitution. Intersection density is not significant, and its sign is not even stable between the five- and eight-city samples, so no conclusion is drawn from it. Districts with more disordered street orientation (higher entropy) are more brittle. That the signal is carried by  $H_0$  rather than

$H_1$  has a concrete planning meaning. In green-dense cities the structure that matters is not isolated “deserts” surrounded by served land but the connectivity of the served regions themselves, and that connectivity is what disruption erodes.

For design, this suggests a *hypothesis* rather than a prescription, and one the present data qualify. One reading—that *path redundancy*, not raw proximity, protects access to nature under stress—was tested directly and not supported: the descriptors associated with resilience (circuitous routing, longer segments) are in fact associated with *less* mesh redundancy, not more. What the metric rewards is a coarser, more stable served configuration under random link loss. Whether that stability is a genuine planning virtue or partly an artefact of how the topological summary scales with configuration granularity is what an independent, validated resilience outcome would have to settle. Any reframing of an SDG 11.7 design brief from “put a park within  $x$  metres” toward “ensure the walk to it survives the loss of individual streets” therefore remains a direction for testing: the association is observational, the outcome is a proposed topological indicator rather than a validated resilience measure, its own proposed mechanism does not survive a direct test, and green-space quality, public accessibility and real hazard exposure are not modelled.

**Limitations.** The caveats are grouped by the kind of claim they qualify.

*Inference and spatial dependence.* The design is cross-sectional and observational, so the regression identifies *associations* between morphology and topological resilience, not causal effects. Districts are also not independent: they nest within eight cities and are spatially contiguous. City fixed effects absorb additive between-city differences, but two residual problems remain. First, eight cities furnish only eight clusters for cross-city inference; clustering the standard errors by city widens the intervals, but all three headline effects still remain significant under this conservative treatment (circuitry  $p = 2 \times 10^{-4}$ , mean street length  $p = 5 \times 10^{-19}$ , orientation entropy  $p = 0.036$ ; Table 3). Eight clusters is nonetheless modest, so greater weight is placed on the estimated spatial models rather than on small-cluster asymptotics. Second, the residuals retain weak positive spatial autocorrelation within most cities (global Moran’s  $I$  up to 0.18; pooled 0.24,  $p < 10^{-100}$ ). This is addressed by estimating spatial-error and spatial-lag models on a within-city weights matrix (Table 3): both absorb strong spatial dependence ( $\hat{\lambda} = 0.53$ ,  $\hat{\rho} = 0.51$ ), yet all three headline descriptors keep their sign and remain significant at  $|z| \geq 5$ . The morphology signal is therefore not an artefact of spatial autocorrelation. Broader replication—more cities, and cities with sparser OpenStreetMap coverage than this set—remains the natural next step.

*Omitted variables.* Morphology is only one of many district attributes that could be associated with resilience. The four that the data support are controlled—green-space area, road hierarchy, relief and city-centre distance (Table 5)—and the headline morphology effects survive their inclusion, so the association is not explained away by these four (which is weaker than ruling them out as causes). But population and built density, land-use mix, park-level size and quality, socioeconomic status and pedestrian-infrastructure quality would require external data (e.g. census and GHSL built-up layers) and remain uncontrolled; city fixed effects remove their between-city component but not their within-city variation, so some residual omitted-variable bias may persist.

*Shared origin of predictor and outcome.* Both the morphology descriptors and the resilience outcome are computed from the *same* street network, so part of the association is structural rather than a clean independent-variable/dependent-variable relationship: a district cannot have, say, low circuitry and a routing topology that behaves as if it were highly circuitous. The regression is therefore read as characterising *which* formal properties co-vary with resilient access, not as an external cause acting on an independent outcome. Two observations temper the concern: the morphology descriptors explain only a modest share of the variance ( $R^2 \approx 0.20$ ), so the outcome is

far from a deterministic re-encoding of the predictors; and the accessibility field also depends on the green-space configuration, which is exogenous to street form. A direct check is available. Because the AUC scales strongly with the raw size of the persistence diagram (its correlation with the district’s total  $H_0$  persistence is +0.83), the morphology signal is refitted while controlling for that same-network measure of diagram magnitude. Circuity and mean street length survive ( $-7.39 \rightarrow -2.88$  and  $-0.050 \rightarrow -0.014$ , both still  $p < 10^{-8}$ ), so they carry configurational information beyond the sheer size of the diagram; orientation entropy, by contrast, loses significance and flips sign ( $+1.85 \rightarrow -0.14$ ,  $p = 0.10$ ), revealing that its apparent effect was largely a re-encoding of diagram magnitude—a concrete instance of the shared-origin risk. This is only a partial mitigation: the control is itself network-derived, and a design that predicted one city’s resilience from another’s morphology, or from an independent disruption dataset, would be needed to break the shared-origin coupling entirely.

*What the metric measures, and how far it is validated.*  $D(\rho)$  is a *proposed proxy* for connected-access resilience, and this is the study’s central limitation. What is established rigorously is an *association* between morphology and this particular Wasserstein/AUC topological indicator; that circuity and mean street length “explain resilience” in a broader sense follows only to the extent that the indicator is itself a valid measure of resilience, which is not fully demonstrated. Evidence of two kinds is offered. Its *construct validity* is supported by its behaviour—it rises with disruption over most of the range (though not monotonically at high  $\rho$ , where finite classes vanish), builds on field-perturbation stability of the diagram distance on the fixed domain [6] (with the domain-change caveat of Section 4), and yields a morphology relationship that survives disruption model, district resolution, denoising threshold, pivot count, spatial-error/lag modelling and four omitted-variable controls. On *criterion* validity, the benchmark against a distance-based accessibility-degradation measure, computed for all eight headline cities so that validation and analysis samples coincide, is positive at the level at which the metric is used but only moderately and heterogeneously so. Within city, the unit the fixed effects identify from, the topological AUC correlates +0.33 ( $p < 10^{-102}$ ) with the benchmark pooled over the 3,999 districts and is positive in seven of the eight cities, yet strong in only one (Amsterdam +0.47) and near zero in most; it remains non-redundant (the morphology signal survives controlling for it). Its *discriminant* validity is cleaner: against connectivity degradation the eight-city correlation is significantly negative ( $\rho = -0.21$ ), which shows the metric is not a restatement of connectivity or percolation robustness but a distinct construct—the graded topological persistence of the *connected* green-access structure under progressive network thinning. That construct has a definite scope: it captures graded access degradation, not total *disconnection* (unreachable nodes leave the finite diagram), so the metric should be paired with a disconnection count for total access loss. An attempt to fold disconnection into a single scalar is reported, and the result is informative. A finite cap ( $\tau = 60$  min) recovers the connectivity sign ( $-0.10 \rightarrow +0.02$  on a two-city check) but erodes the graded-distance correlation ( $+0.41 \rightarrow +0.06$ ), and a principled extended-persistence diagram, which gives the disconnected components finite coordinates in a separate channel, behaves the same way on Amsterdam (connectivity +0.31 but distance only +0.05), because the high-persistence component classes dominate a single Wasserstein match. Graded resilience and disconnection are, then, two channels that a single scalar conflates; reporting them separately, as done here, is the right treatment. A joint multi-channel metric, and a full head-to-head against an externally *validated* criterion (a formal two-step floating catchment accessibility model, or observed post-disruption service loss) rather than the convergent proxies benchmarked here, is the natural next step. Two further caveats: AUC depends on network and district size and on the number of  $H_0$  classes, so it is comparable across cities only in relative ranking, not in absolute level (city fixed effects absorb this scale in the regression), and  $\rho^*$  depends on the chosen disruption range. The down-turn of the İstanbul and Phoenix curves at  $\rho = 0.5$  reflects

finite nodes dropping out of the diagram under near-total fragmentation, not recovered access, and marks the upper edge of the metric’s useful range. Green spaces are treated as equivalent point sources at their centroids, ignoring size, quality and public/private status; a large park reduced to its centroid can misplace the true walking distance to its entrances, and the sensitivity of the results to multi-entrance or area-based access points is not tested here. OSM mapping completeness, pedestrian-network coverage, green-space classification and topological conventions also vary across countries and are not quantified, so data-quality heterogeneity is a genuine external-validity limit, and one the city fixed effects do *not* remove: fixed effects absorb each city’s unobserved mean level, not city-specific differences in data quality or in the morphology–resilience slope itself. The per-city coefficients (Table 4) and the leave-one-city-out transfer above are what speak to slope stability; absolute cross-city comparison of coefficients is not warranted. The class-exclusion sensitivity below addresses the tag-mix concern but not these geometric and completeness ones.

*Disruption and data.* Random removal is spatially uniform and does not reproduce the clustered failures of a real flood or earthquake; targeted removal addresses criticality but not spatial correlation. The real 100-year flood scenario added above makes this concrete: it materially affects only Vienna and does not carry the distributed-disruption morphology signal, showing that resilience to a localised hazard is a genuinely different quantity—one this paper does not claim to explain. Earthquake and multi-hazard disruption models are left to future work. Targeted betweenness is approximated from  $k=500$  sampled pivots, as exact  $O(V|E|)$  betweenness is intractable at city scale; the ranking is stable in the pivot count: on the Amsterdam network, halving or doubling  $k$  (to 250 or 1,000) leaves the edge ranking essentially unchanged (Spearman  $\rho = 0.985$  against  $k=500$ , with 77%/90% overlap of the top-decile edges)—so the targeted scenario is not an artefact of the pivot count. Results use a persistence-thresholded  $H_0$  metric, so reported magnitudes are denoised estimates; varying the threshold over a  $4\times$  range (0.5–2.0 walk-minutes) preserves every coefficient sign and keeps the three headline descriptors significant at  $p < 10^{-6}$ , and the full disruption grid confirms the signs, significances, ranking and  $\rho^*$ . OSM completeness varies by city and is not quantified here, and the network-distance Vietoris–Rips variant of the filtration is left to future work.

*Modelling choices.* Two inputs could in principle shape the results: the inclusive green-space definition and the constant 4.8 km/h walking speed. Both were tested. The strict, accessibility-conservative green-space subset leaves the headline signs and significance unchanged (Section 6); and, because speed enters only as a global rescaling of the field, the walking-speed choice cannot alter the coefficient signs, significances or the city ranking, which is confirmed empirically ( $\rho > 0.98$  between the 4.8 and 4.0/5.0 km/h district AUCs). These are therefore not load-bearing assumptions.

Finally, “resilience” here is a *structural* property—the robustness of the access topology to synthetic edge removal, measured on a single cross-sectional snapshot per city—not a temporal recovery process; how access rebuilds over time after a real disruption, and the term should be read in that narrower, topological sense.

Read together, these caveats narrow the claim: within these eight cities, and under the tested *synthetic, distributed* edge-removal disruption (random and targeted), street-network morphology is a consistent descriptive correlate of green-space access resilience—robust in sign across disruption models, district resolutions, spatial specifications and the added controls—rather than a causally identified or cross-city-generalisable driver. Crucially, this scope is *empirical*, not merely cautionary: under the real, spatially *clustered* 100-year flood scenario the pooled morphology effects do not reproduce (Section 6). Distributed removal thins links roughly evenly, so the descriptors that govern how a district’s served  $H_0$  structure reorganises under diffuse loss are what the metric responds to; a clustered flood instead removes a spatially contiguous block of the network at once, a regime in which local morphology is dominated by *where* the hazard falls rather than by routing form. The general phrasing “street morphology predicts urban resilience” therefore overreaches on

two counts—“urban resilience” (the metric measures connected-access resilience, not global urban resilience) and “predicts” (the relationship is associational); the defensible statement is that street morphology is associated with connected-access resilience *under the tested synthetic, distributed disruption scenarios*, and its transfer to real clustered hazards is not established.

## 8 Conclusion

This study defines a topological metric for the resilience of green-space access, the Wasserstein degradation curve of persistent-homology diagrams under network disruption, and showed across 3,999 districts in eight contrasting cities on four continents that, under synthetic distributed disruption, street-network morphology is associated with how robustly a city’s pedestrian network keeps its *connected* served locations close to nature. Two effects are robust throughout—longer street segments (the largest) and more circuitous routing are associated with greater resilience, and both survive a wild cluster bootstrap and spatial-error and spatial-lag models—while higher orientation entropy is associated with less resilience but proves the least dependable, weakening across cities and vanishing once diagram magnitude is controlled. These associations hold for distributed disruption and do not carry to the real clustered-flood scenario. The contribution has two boundaries. The relationship is an *association*, not a causal effect: the design is cross-sectional and observational. And it is an association with the *proposed* topological indicator; whether that indicator is a fully valid measure of resilience—rather than a well-behaved, configuration-sensitive proxy—is the study’s central open question. Eight cities also still furnish only eight clusters (see Limitations). The paper is precise about what the metric captures: the graded topological persistence of the *connected* green-access structure under progressive network thinning—moderately and heterogeneously validated within city against a distance-based accessibility benchmark across all eight cities, and shown by its negative correlation with connectivity degradation to be a distinct construct from percolation robustness rather than total accessibility including outright disconnection. The mathematics of persistence turns a static SDG 11.7 indicator into a dynamic, form-aware one. Future work will extend the real-hazard analysis beyond river floods to earthquakes and multi-hazard scenarios, and add a network-metric Vietoris–Rips filtration, toward a resilience-aware accessibility standard for sustainable cities.

**Reproducibility.** All data are open (OpenStreetMap, GHS-UCDB) and the pipeline is open-source Python (OSMnx, NetworkX, GeoPandas, GUDHI/**persim**, H3, **statsmodels**); every reported number is recorded in the project’s results log and traces to a committed artifact.

## A Robustness and sensitivity analyses

This appendix collects the diagnostics summarised in the Results. Throughout, the three headline descriptors are circuitry, mean street length and orientation entropy.

**Spatial dependence.** Districts nest within cities and are spatially contiguous, so the fixed-effects OLS residuals are not independent. Global Moran’s  $I$  on first-ring H3 adjacency is positive and significant in every city (up to 0.18 in Barcelona and Singapore, mostly permutation  $p \leq 0.01$ ), and Anselin Lagrange-multiplier tests on the OLS residuals decisively reject the non-spatial model (LM-error  $p < 10^{-135}$ , LM-lag  $p < 10^{-177}$ ; the robust LM statistics favour the lag form, 227 vs. 31), with both spatial models cutting the AIC by more than 500 relative to OLS (22,063  $\rightarrow$  21,475 for the

Table 3: Inference robustness of the eight-city district regression: city-clustered standard errors (eight clusters, the most conservative treatment of the nesting) and estimated spatial-error and spatial-lag models (ML, within-city H3 weights). Coefficients are on AUC. All three headline descriptors survive city clustering and both spatial models, despite substantial estimated spatial dependence.

| Morphology descriptor | coef (OLS) | $p$ (cluster)             | Sp. error $z$          | Sp. lag $z$         |
|-----------------------|------------|---------------------------|------------------------|---------------------|
| circuitry             | -7.39      | $2.2 \times 10^{-4}$ ***  | -5.2                   | -6.6                |
| mean street length    | -0.050     | $4.5 \times 10^{-19}$ *** | -7.5                   | -8.8                |
| orientation entropy   | +1.85      | 0.036 *                   | +10.1                  | +9.6                |
| intersection density  | +0.02      | 0.89                      | -1.5                   | -1.2                |
| spatial parameter     |            |                           | $\hat{\lambda} = 0.53$ | $\hat{\rho} = 0.51$ |

error model, 21,376 for the lag model), so spatial-error and spatial-lag models are estimated spatial-error and spatial-lag models (Table 3). Both absorb strong dependence ( $\hat{\lambda} = 0.53$ ,  $\hat{\rho} = 0.51$ ), yet the three headline descriptors keep their sign and stay strongly significant ( $|z| \geq 5$ ). City-clustered standard errors (Table 3), the most conservative treatment of the eight-cluster design, leave all three significant, including orientation entropy.

**Multicollinearity and cross-city heterogeneity.** The headline descriptors are not collinear: within-city, the largest pairwise correlation among circuitry, mean street length and orientation entropy is 0.31, and all variance-inflation factors are below three (the only strong pair is mean street length and intersection density,  $r = -0.80$ , which is why intersection density is unstable and not relied on). The pooled slopes are also not a pooling artefact: fitting the regression separately per city, each headline sign holds in seven of the eight cities (Table 4), with Nairobi the exception for circuitry and Phoenix for orientation entropy. The magnitudes, however, are *not* constant across cities, and this is stated explicitly rather than assumed away. A joint test of the morphology  $\times$  city interactions rejects slope homogeneity ( $F = 12.5$  on 21 terms,  $p < 10^{-41}$ ), and the per-city slopes are widely dispersed (for circuitry and mean street length the between-city standard deviation is of the same order as the mean slope; coefficient of variation  $\approx 1.0$ ). A random-slopes mixed model (standardised) agrees: the mean slope and its between-city SD are  $-0.31 \pm 0.17$  for mean street length,  $+0.28 \pm 0.21$  for orientation entropy and  $-0.14 \pm 0.13$  for circuitry—directionally consistent but with real magnitude heterogeneity. City fixed effects remove only mean-level differences, not this variation in slope. The pooled coefficient is therefore interpreted throughout as the *average within-city conditional association across these eight cities*, not as a homogeneous or universal effect of morphology on resilience; mean street length is the most stable (largest mean-to-SD ratio), circuitry the least.

**Out-of-sample cross-city transfer (leave-one-city-out).** With only eight clusters, the sharper question is whether the relationship *transfers* to a city the model never saw. A leave-one-city-out test addresses this directly: within-city slopes are fitted on seven cities and used to predict the ordering of districts in the held-out eighth (the held-out city’s own mean is removed, so the target is its within-city ranking, which needs no estimated fixed effect). The transfer is positive in *all eight* held-out cities, with mean out-of-sample Spearman +0.42 (Barcelona +0.61, Singapore +0.58, Vienna +0.49, İstanbul +0.47, Nairobi +0.39, Amsterdam +0.36, Bogotá +0.34, and Phoenix the weak case at +0.07,  $p = 0.02$ ). Morphology fitted elsewhere therefore predicts, imperfectly but con-



Table 4: Per-city regression coefficients (AUC on morphology, one city at a time). Each headline sign—circuitry (−), mean street length (−), orientation entropy (+)—holds in seven of the eight cities.

| City      | circuitry | mean street length | orientation entropy |
|-----------|-----------|--------------------|---------------------|
| Amsterdam | −4.07     | −0.04              | +1.17               |
| Barcelona | −6.57     | +0.02              | +6.17               |
| Bogotá    | −18.62    | −0.21              | +1.46               |
| İstanbul  | −14.51    | −0.03              | +3.59               |
| Nairobi   | +4.86     | −0.05              | +2.56               |
| Phoenix   | −3.70     | −0.06              | −0.72               |
| Singapore | −4.12     | −0.04              | +4.58               |
| Vienna    | −5.56     | −0.04              | +3.39               |

sistently, the resilience ranking of an unseen city’s districts—the strongest external-validity evidence the eight-city design can offer, short of an independent validation sample.

**Disruption scenarios (five-city core).** Under *targeted* disruption (removal of high-betweenness edges, ranked by  $k=500$ -sampled betweenness) every coefficient sign in Table 2 is reproduced: circuitry  $-5.77$  ( $p = 4.6 \times 10^{-10}$ ), mean street length  $-0.034$  ( $p = 2.7 \times 10^{-13}$ ), orientation entropy  $+0.50$  ( $p = 1.6 \times 10^{-3}$ ), and intersection density (not significant under the full random grid) reaching  $p = 0.022$  ( $-0.135$ ). Absolute AUC values are lower (Amsterdam mean AUC 2.03 vs. 4.46 under random) but the ranking is unchanged.

**Real flood-hazard scenario.** The earlier low-elevation proxy with a *real* 100-year river-flood scenario, using the open JRC Global Flood Hazard maps [8] (floodMapGL, water depth per  $\approx 1$  km cell): a node is flood-exposed if it lies in a cell with  $\geq 0.5$  m of inundation, and the hazard removal progressively cuts flood-exposed edges. Two honest findings follow. First, real 100-year exposure among these urban centres is strongly city-specific: it is material only in Vienna (23% of nodes, on the Danube floodplain) and negligible elsewhere (Phoenix 0.7%, Amsterdam 0.4%, Barcelona 0.3%; İstanbul, Bogotá, Singapore and Nairobi essentially 0%, their centres sitting off the modelled river-flood plain). Second, where the scenario bites (Vienna), the morphology signal established for *distributed* disruption does *not* cleanly carry over to this *spatially clustered* flood: in a pooled flood-scenario regression the morphology coefficients are non-significant, and within Vienna only circuitry keeps a weak same-signed effect (Spearman  $-0.13$ ,  $p = 0.006$ ) while orientation entropy even reverses. This is consistent with, and concretely demonstrates, the caveat that resilience to network-wide disruption is a different quantity from resilience to a localised, geographically concentrated hazard; morphology governs the former, not necessarily the latter.

**Disruption-grid and resolution sensitivity (five-city core).** A coarser  $5 \times 3$  disruption grid leaves every sign unchanged and keeps the three headline descriptors significant at  $p < 10^{-3}$  (circuitry  $-8.99$ , mean street length  $-0.0478$ , orientation entropy  $+1.17$ ), with the city AUC ranking and  $\rho^*$  values within  $\approx 0.02$  of the full  $8 \times 10$  grid. The one descriptor that moves is intersection density: significant on the coarse grid ( $-0.199$ ,  $p = 8.6 \times 10^{-3}$ ) but not on the full grid ( $p = 0.22$ ). Re-tiling at H3 resolutions 7 (485 districts) and 9 (13,782 districts) likewise leaves the three headline descriptors significant at  $p < 10^{-3}$  with constant signs across a  $28\times$  range of district counts; magnitudes scale with cell size but no sign flips. Figure 5 summarises both axes.

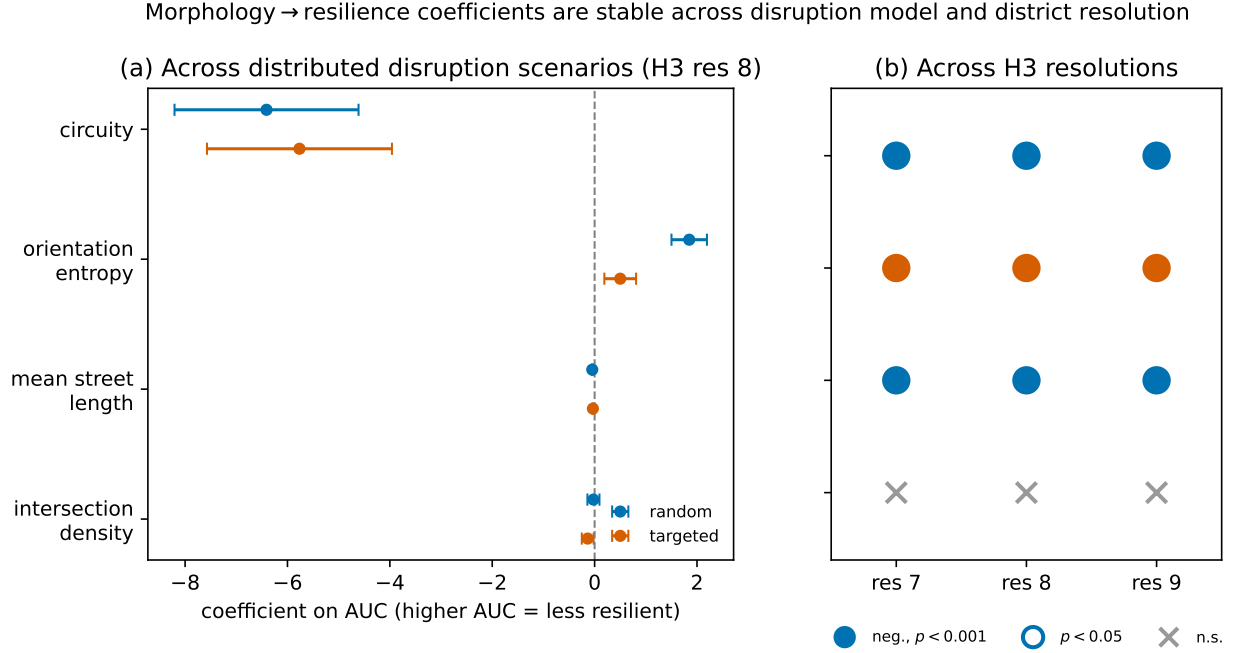


Figure 5: Robustness of the morphology↔resilience relationship. **(a)** Regression coefficients (with 95% confidence intervals) for each morphology descriptor under the two distributed disruption scenarios (random and targeted) at H3 resolution 8; the coefficient signs agree, and the three headline descriptors (circuitry, mean street length, orientation entropy) stay significant, while intersection density crosses zero under random disruption. **(b)** Sign and significance of each coefficient across H3 resolutions 7/8/9: the three headline descriptors stay significant with a constant sign, while intersection density, the weakest and grid-sensitive effect, is not significant at the headline full-grid specification.

**Persistence-threshold sensitivity (five-city core).** Varying the  $H_0$  denoising threshold over a  $4\times$  range leaves the headline unchanged: at 0.5, 1.0 and 2.0 walk-minutes the three descriptors keep their sign and stay significant at  $p < 10^{-6}$  (circuitry  $-11.20/-10.16/-8.03$ ; mean street length  $-0.050/-0.048/-0.042$ ; orientation entropy  $+1.10/+1.01/+0.93$ ). Magnitudes shrink smoothly as more denoising is applied but never change sign, and intersection density stays weak (ns at 0.5 and 1.0,  $p = 0.04$  at 2.0). The betweenness pivot count is equally inconsequential: on the Amsterdam network, halving or doubling  $k$  (to 250 or 1,000) leaves the edge ranking essentially unchanged (Spearman  $\rho = 0.985$  against  $k=500$ , with 77%/90% overlap of the top-decile edges).

**Omitted-variable controls (five-city core).** City fixed effects absorb only between-city confounds, so the district-level controls that the data support are added—total green-space area, the share of edges on the trunk/primary/secondary/tertiary road hierarchy, within-district relief (elevation standard deviation), and distance to the urban-centre centroid—and refit (Table 5). Three of the four controls are themselves significant (more green area and greater centre distance with greater resilience, more major-road edges with less), yet the three headline descriptors keep their sign and stay significant at  $p < 10^{-11}$  (circuitry  $-9.55$ , mean street length  $-0.039$ , orientation entropy  $+1.11$ ). Population and built density, land-use mix, socioeconomic status and pedestrian-infrastructure quality require external data and remain uncontrolled (Limitations).

Table 5: Omitted-variable robustness (five-city core): district regression before and after adding four controls,  $n = 2,603$ , with city fixed effects. The three headline morphology effects keep their sign and significance; the added controls explain additional variance ( $R^2$  0.17  $\rightarrow$  0.21). Sig.: \*\*\* $p < 0.001$ , \*\* $p < 0.01$ .

| Variable                            | base<br>coef ( $p$ ) | +controls<br>coef ( $p$ ) |
|-------------------------------------|----------------------|---------------------------|
| circuitry                           | −10.16 ***           | −9.55 ***                 |
| mean street length                  | −0.048 ***           | −0.039 ***                |
| orientation entropy                 | +1.01 ***            | +1.11 ***                 |
| intersection density                | −0.09                | −0.10                     |
| green-space area (km <sup>2</sup> ) | —                    | −2.25 **                  |
| major-road share                    | —                    | +4.21 ***                 |
| relief (m)                          | —                    | −0.008                    |
| centre distance (km)                | —                    | −0.107 ***                |

## Declarations

**Funding.** The author received no specific funding for this work.

**Competing interests.** The author declares no competing interests.

**Data availability.** All input data are open and publicly available. Pedestrian networks and green spaces were obtained from OpenStreetMap via OSMnx; urban-centre boundaries from the GHS Urban Centre Database (GHS-UCDB R2019A, European Commission Joint Research Centre); residential population from the GHS-POP grid (R2023A, 1 km) for the population-weighted validation; 100-year river-flood depth from the open JRC Global Flood Hazard maps (floodMapGL) for the real-flood scenario; and elevation data from the public AWS Terrain Tiles service. No new data were collected. The derived artifacts underlying every reported figure and table (per-district resilience and morphology tables, regression outputs, and city-level resilience curves) are recorded in the project’s results log, released with the code at <https://github.com/navigaai/nnj-sdg11-topology>.

**Code availability.** The complete analysis pipeline is implemented in open-source Python (OSMnx, NetworkX, GeoPandas, GUDHI, `persim`, H3, `statsmodels`) and is publicly available at <https://github.com/navigaai/nnj-sdg11-topology>, together with the configuration files and the results log that reproduce every reported number.

**Author contributions.** S. Emekci (ORCID 0000-0002-5470-6485) conceived the study, developed the method and software, performed the analysis, and wrote the manuscript.

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